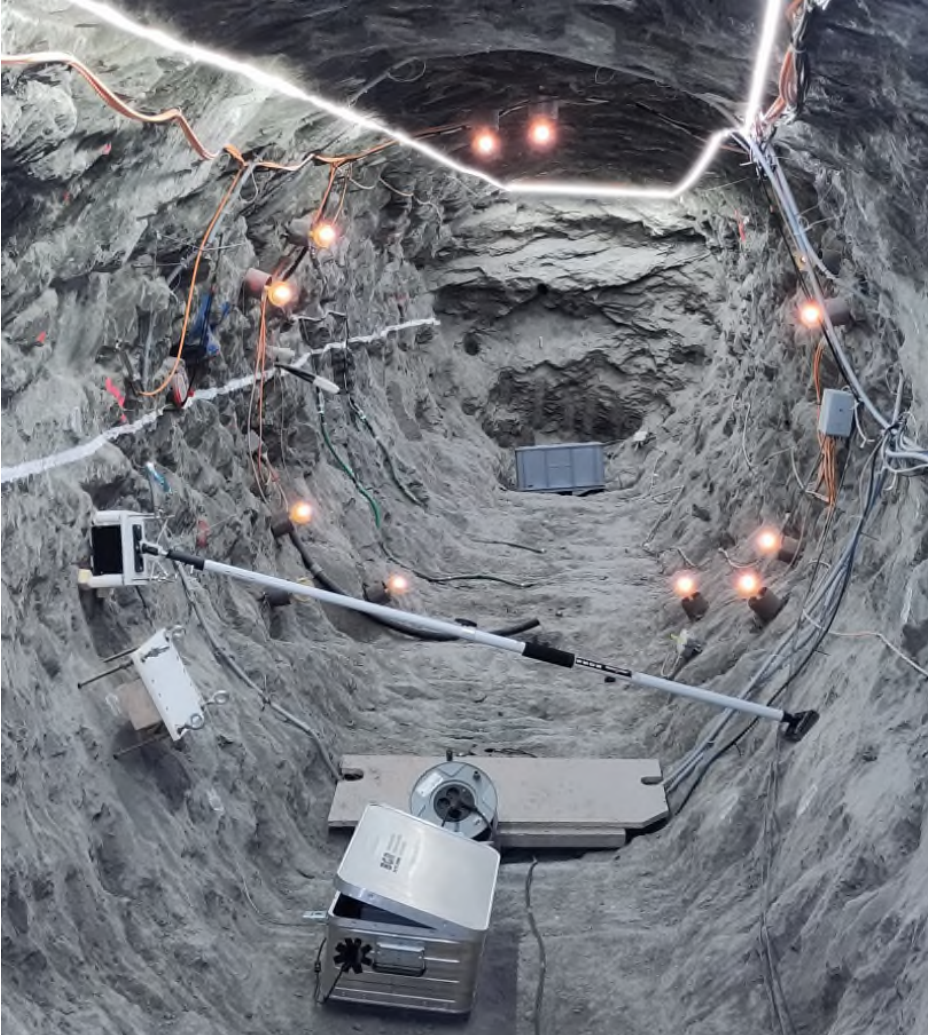
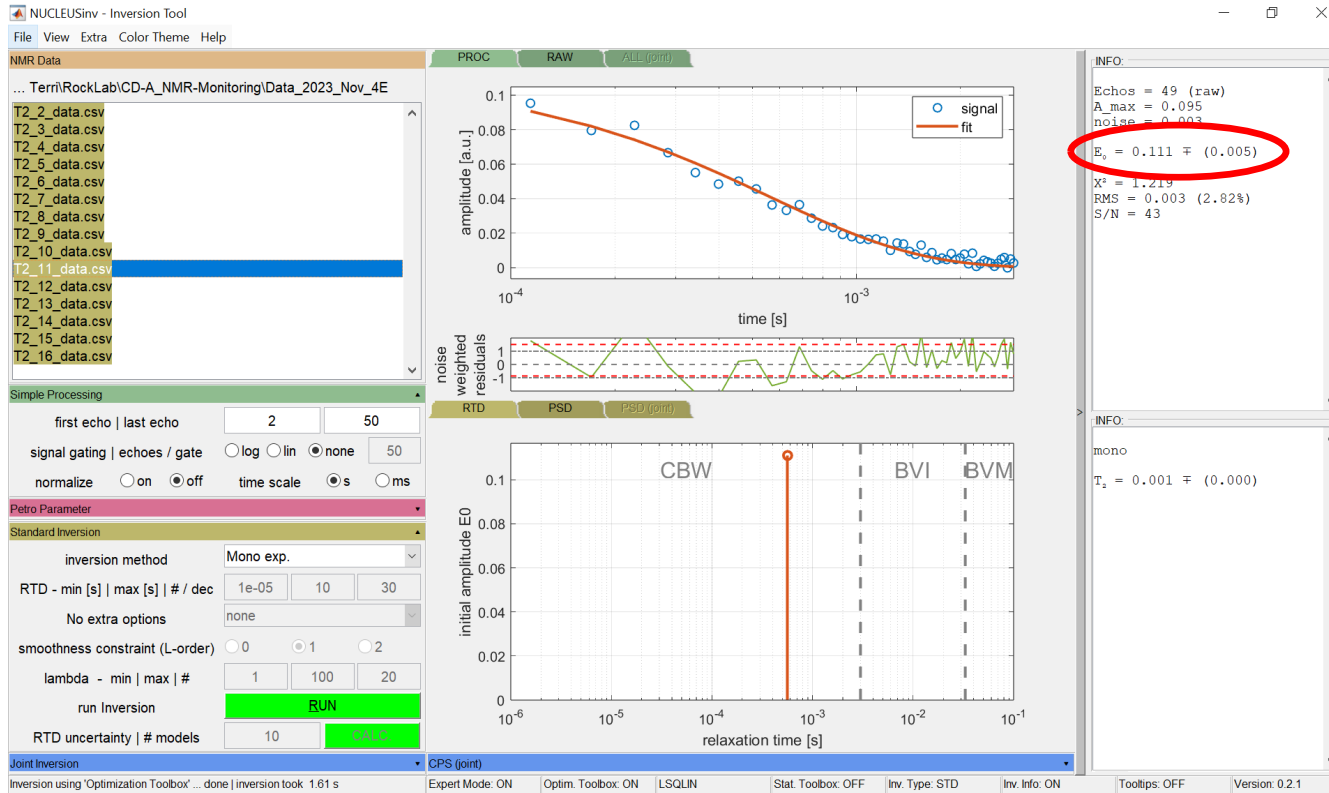


Single-Sided Nuclear Magnetic Resonance on the walls in the CD-A Niches

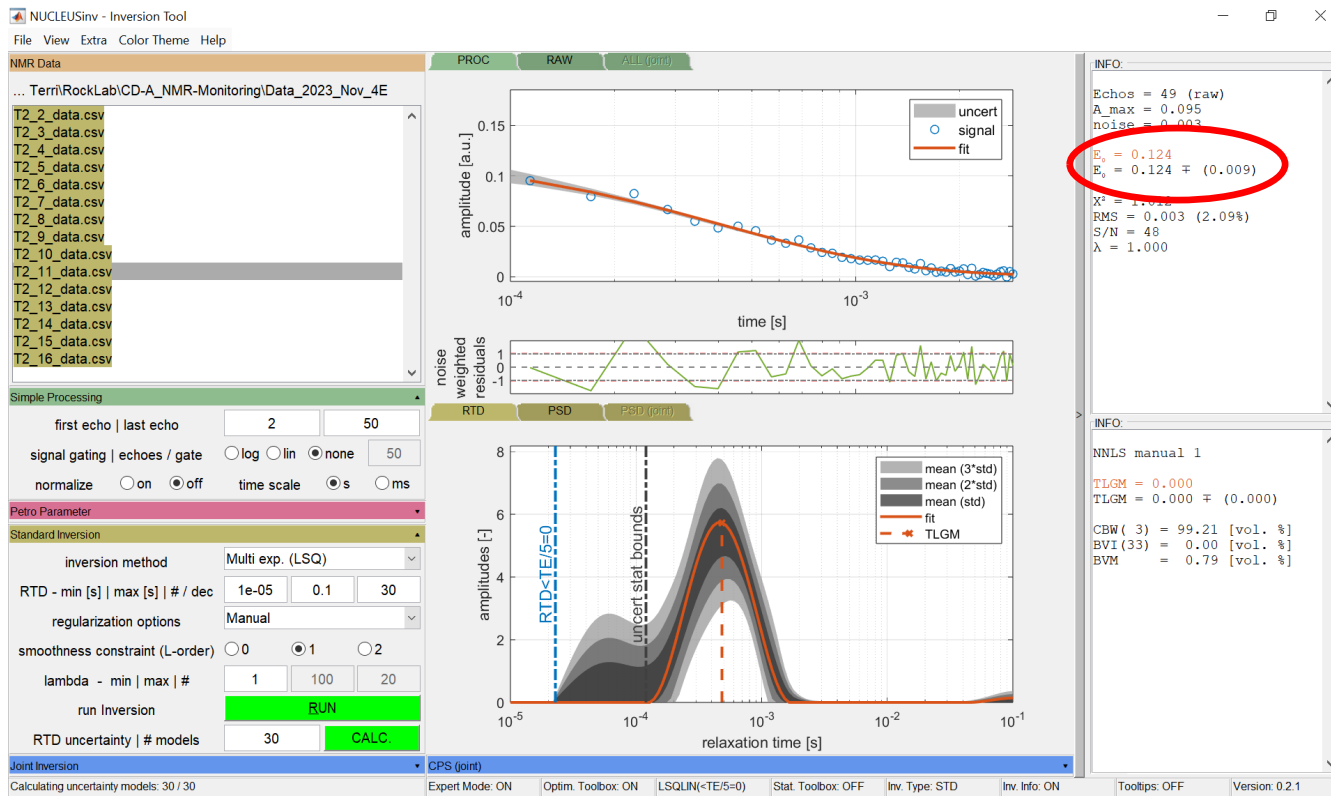
Stephan Costabel, BGR, working group B2.1



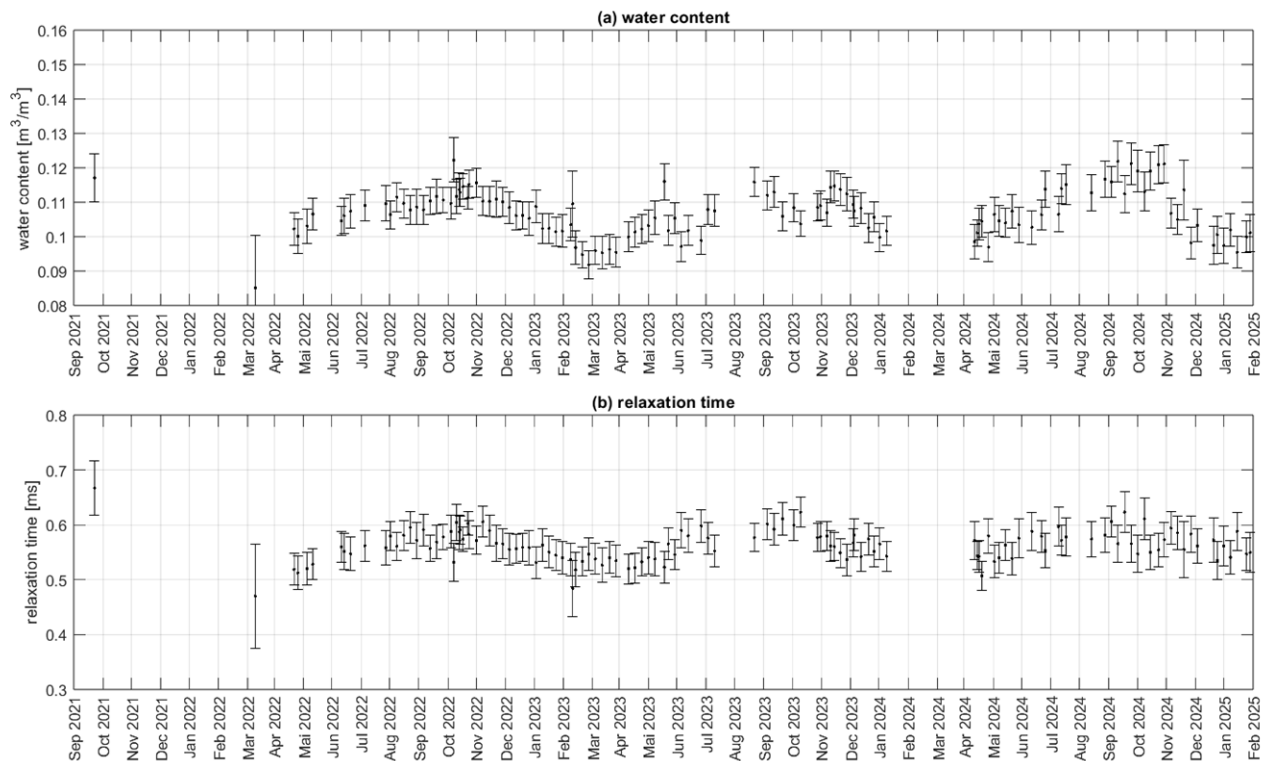
Weekly measurements in open niche - reprocessing of data



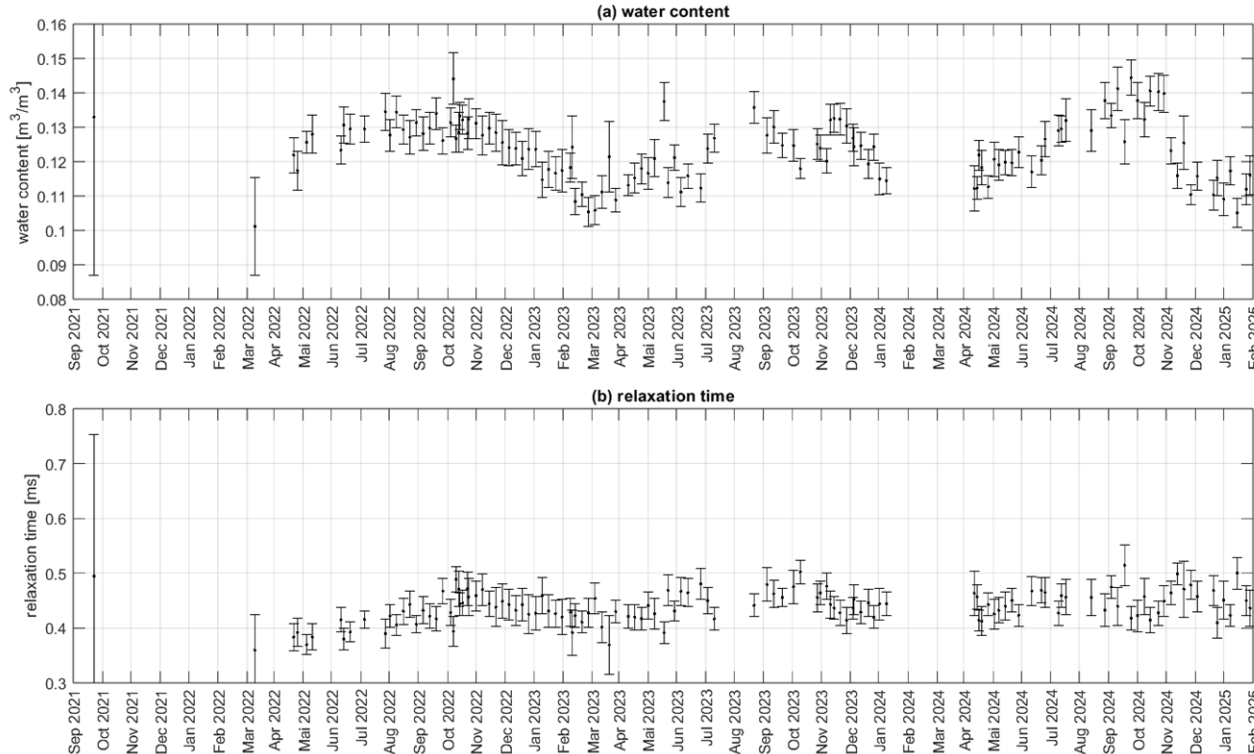
Weekly measurements in open niche - reprocessing of data



Weekly measurements in open niche - data until Feb 2025



Weekly measurements in open niche - data until Feb 2025

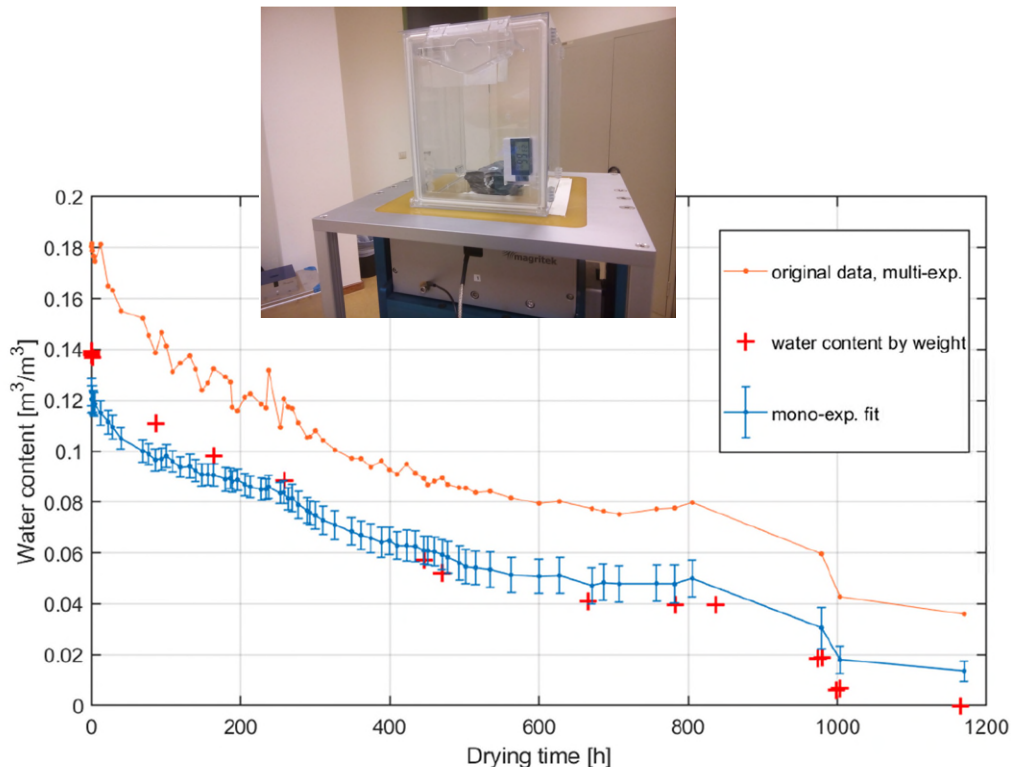


Findings regarding multi-exponential approach:

- Confidence intervals similar to mono-exp. approach
- Water content estimates higher than mono-exp. by approx. 2 % (interlayer water!?)
- Water content estimates with slightly larger difference between minimum and maximum (opposite for relaxation time)

Drying experiments in laboratory

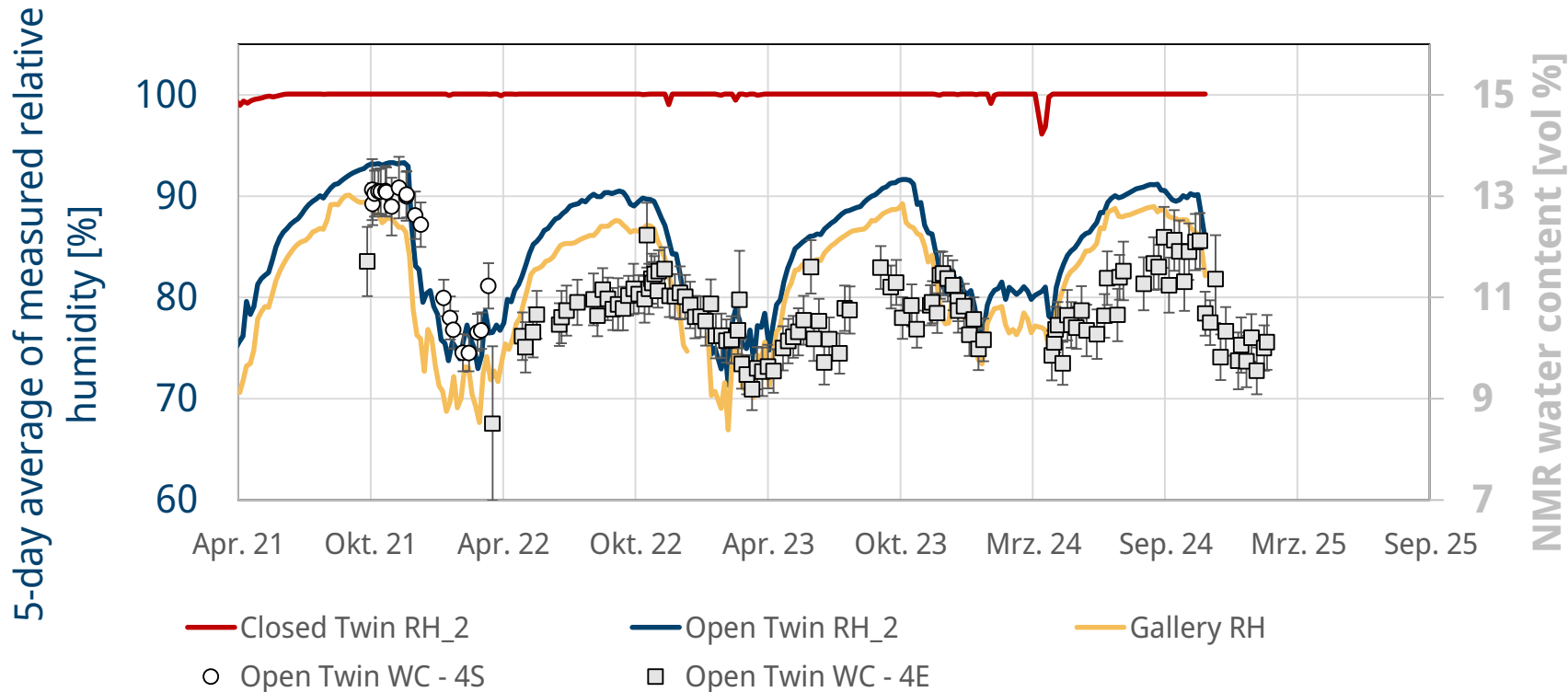
Costabel (2023) Technical Note, MR Experiment in Mont Terri



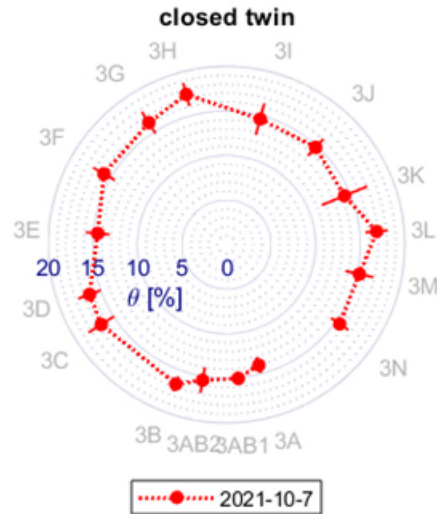
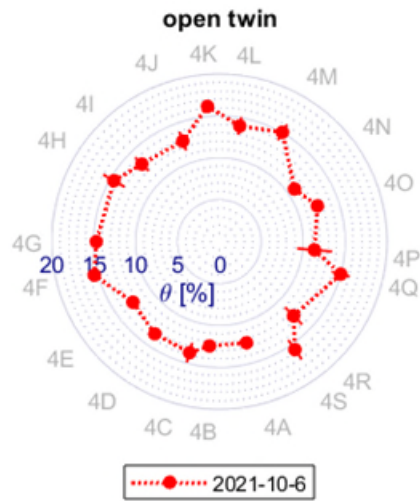
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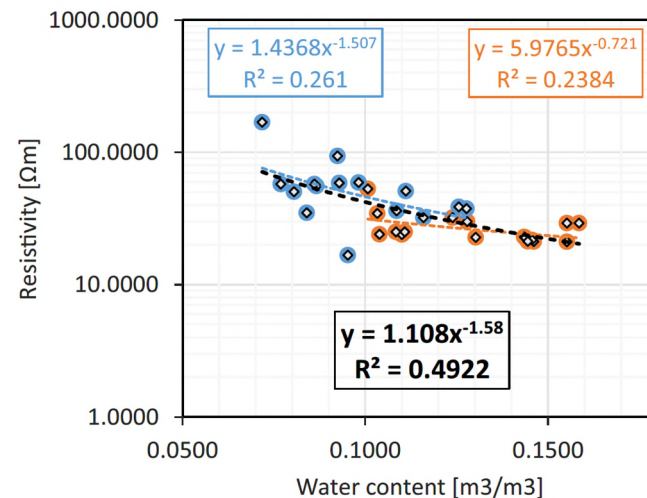
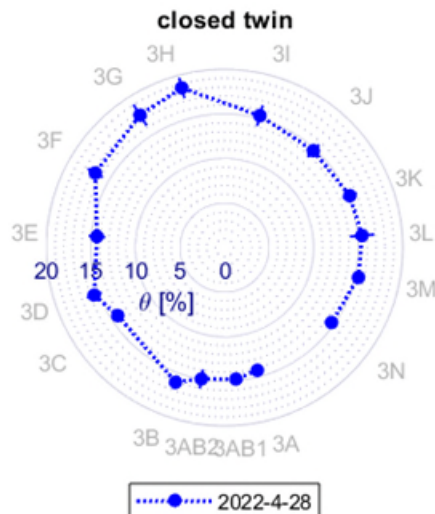
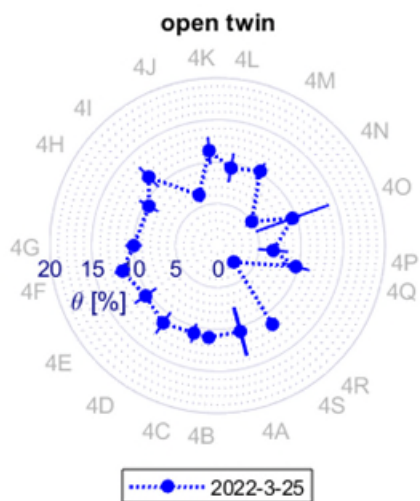
Weekly measurements in open niche - relationship with rel. humidity (apparent phase shift!?)



Seasonal measurements in both niches - data until Feb 2025

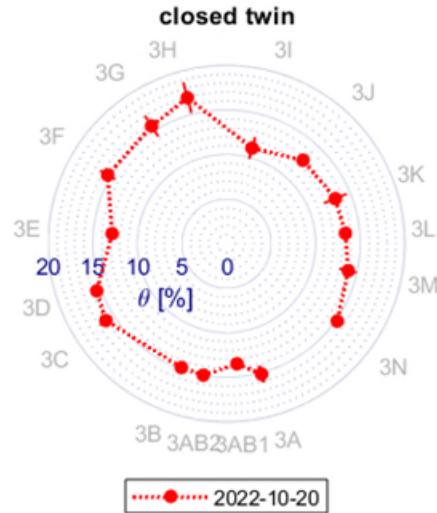
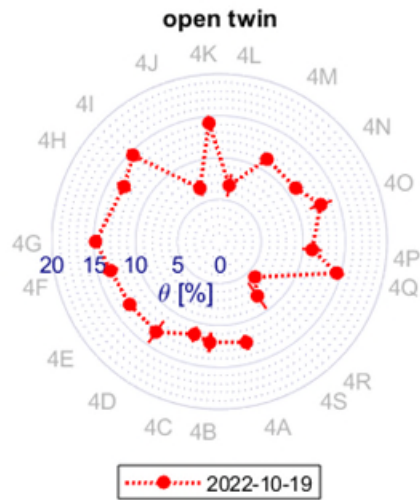


Seasonal measurements in both niches - data until Feb 2025

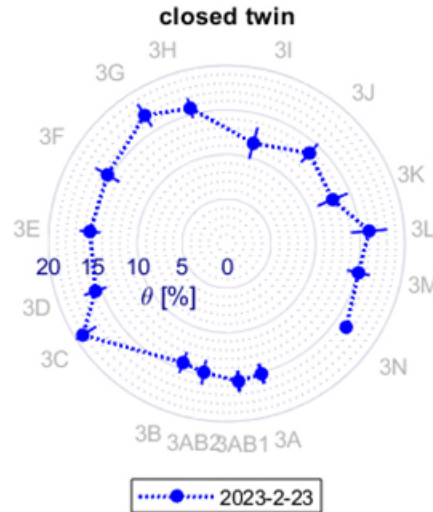
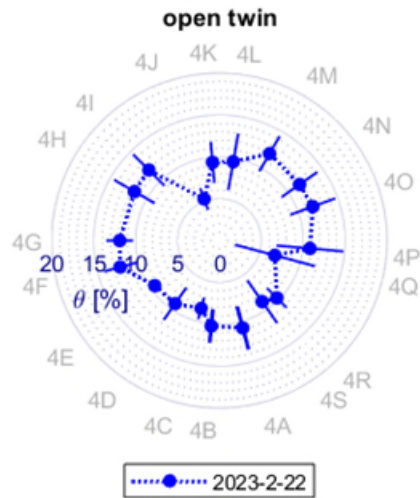


- Closed twin
- Open twin
- ◇ All
- Fit (Closed niche)
- Fit (Open niche)
- Fit (All)

Seasonal measurements in both niches - data until Feb 2025



Seasonal measurements in both niches - data until Feb 2025

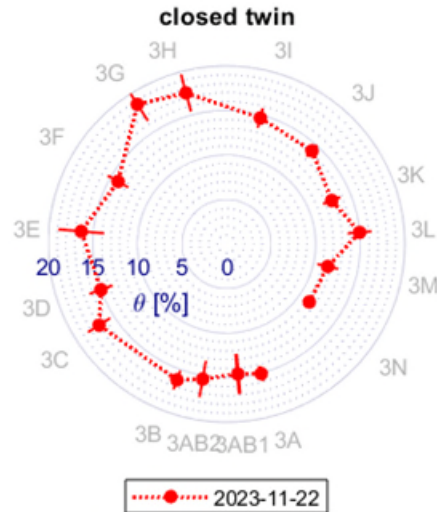
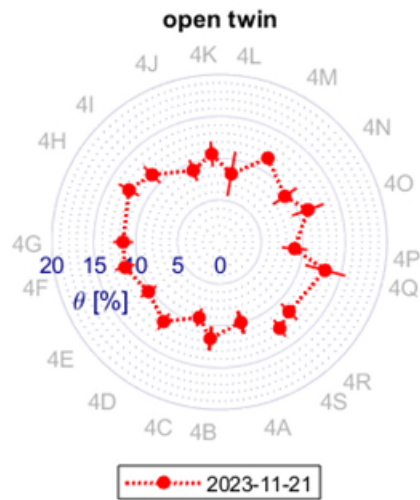


!!?

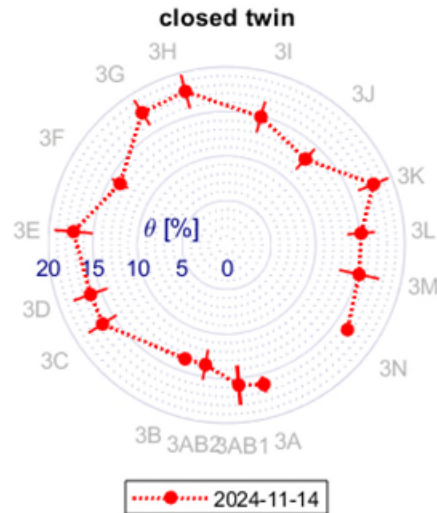
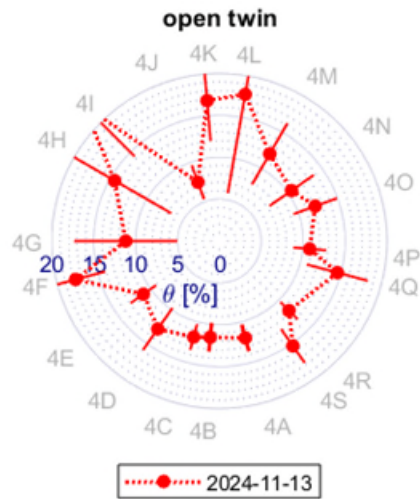
Increase of
electromagnetic noise
level in the niches

!!?

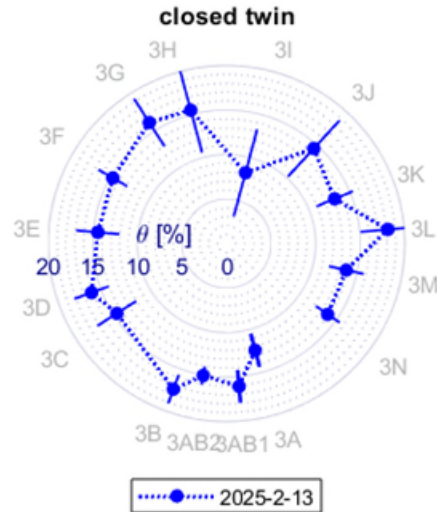
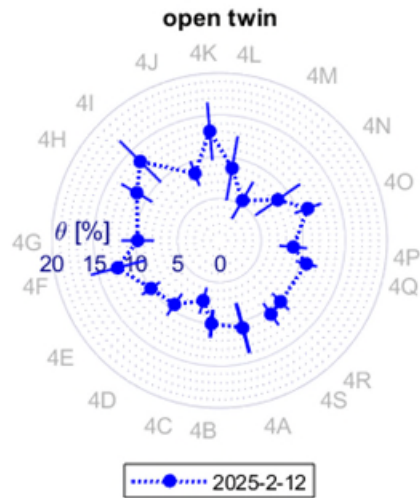
Seasonal measurements in both niches - data until Feb 2025



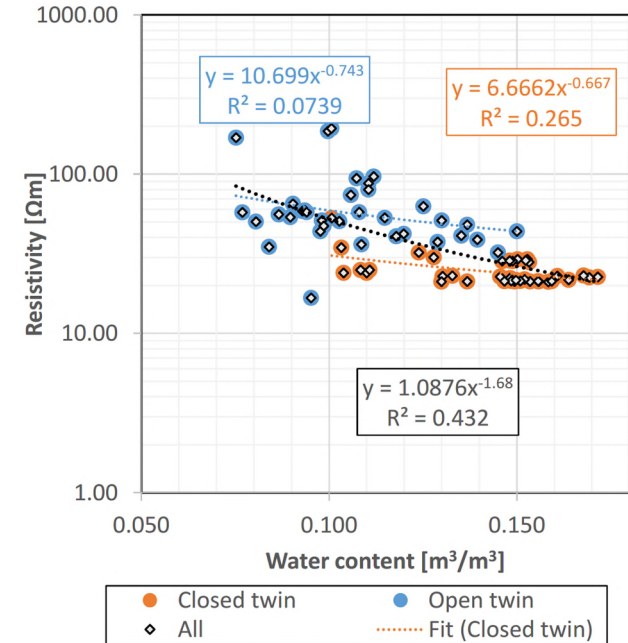
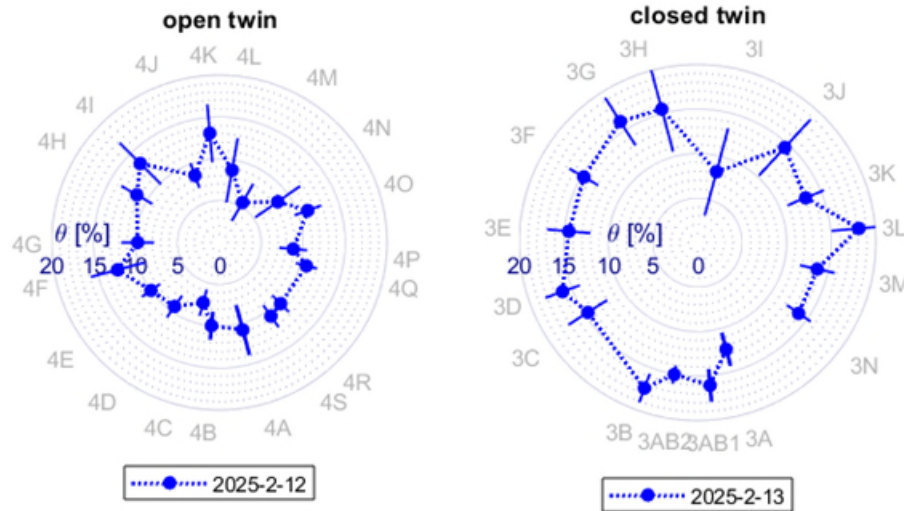
Seasonal measurements in both niches - data until Feb 2025



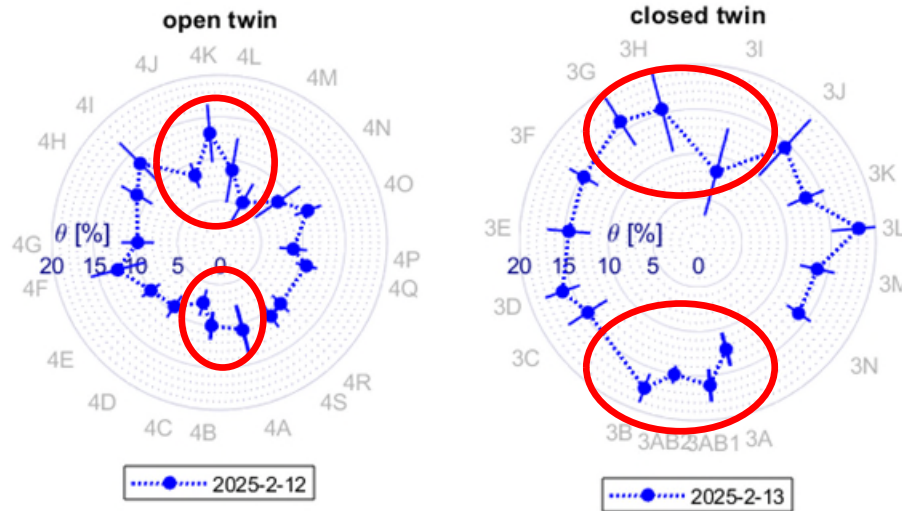
Seasonal measurements in both niches - data until Feb 2025



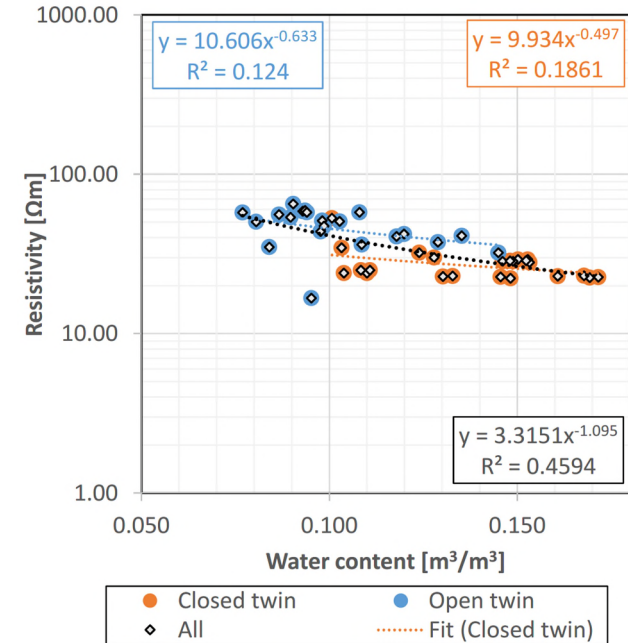
Seasonal measurements in both niches - correlation with ERT



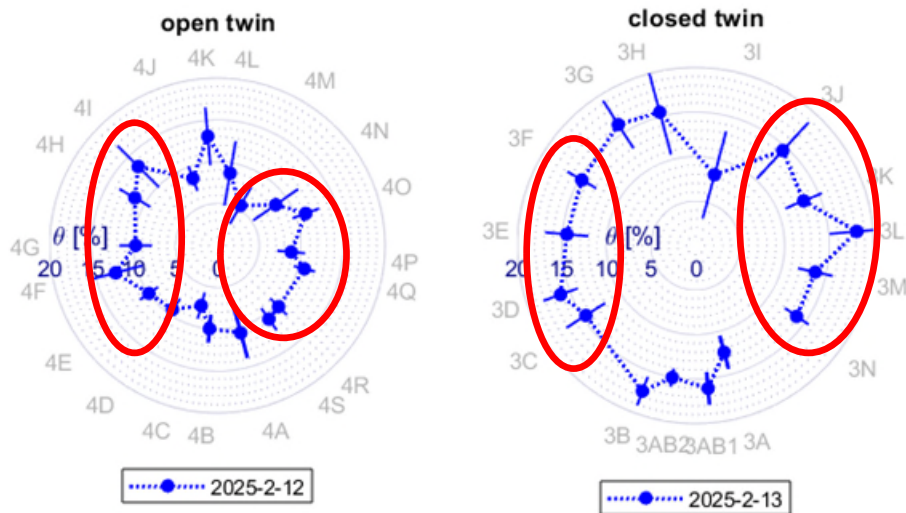
Seasonal measurements in both niches - correlation with ERT



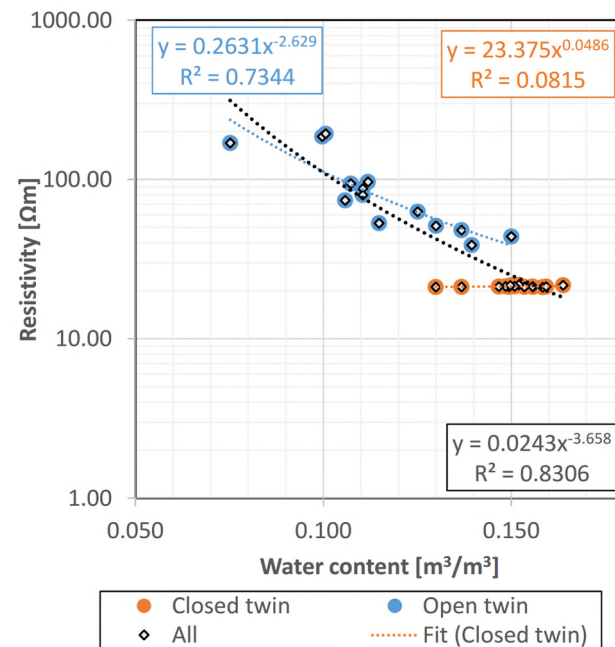
Only top and bottom



Seasonal measurements in both niches - correlation with ERT

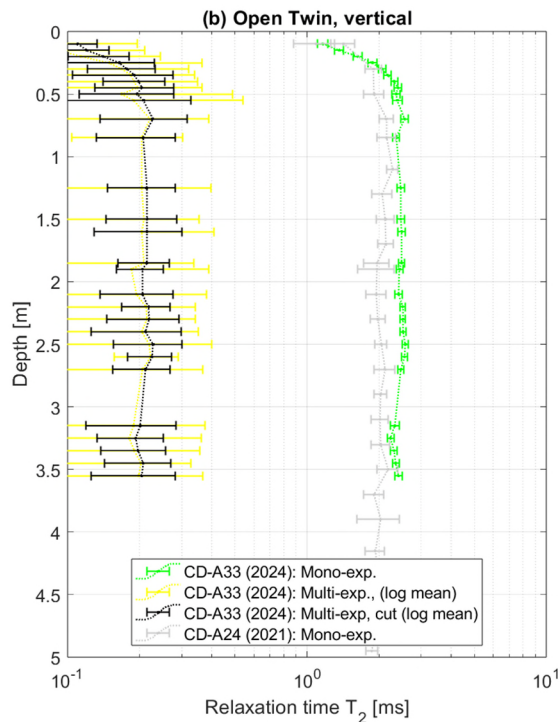
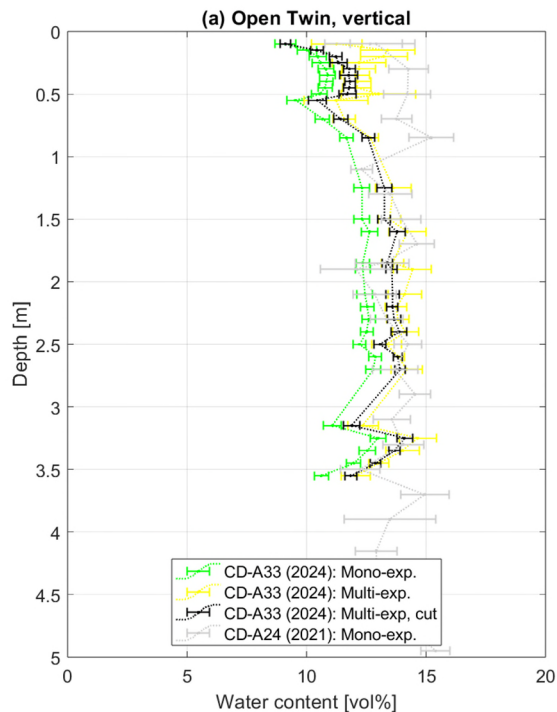


Only sidewalls



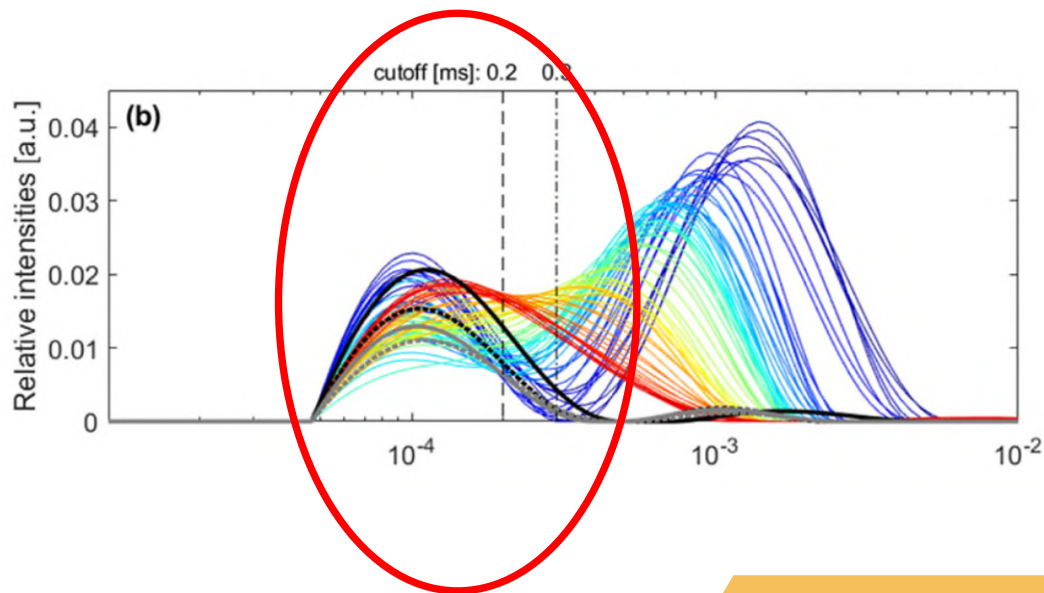
Outlook for 2026 (and rest of 2025)

- Comparison of NMR core scans of 2024 with borehole ERT and other petrophysical data



Outlook for 2026 (and rest of 2025)

- Further laboratory experiments are underway to help interpreting relaxation times:
 - Different well-defined clay mineral powders + grinded OPALINUS clay samples
 - Drying and wetting circles in a climate chamber => NMR at each moisture step
 - Most important question: Do we see the interlayer water in the Opalinus clay?



**Thank you
for your attention!**